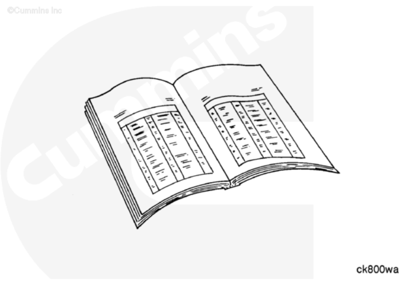
Trainings

Preparatory Steps

Automotive with CM870

- Remove the turbocharger and inspect for reuse. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)



Note : This repair requires the use of different turbocharger housing kits, depending on the engine CPL, and turbocharger serial number (found on the turbocharger dataplate). Reference the appropriate parts release information to determine which turbocharger housing kit to use.

Preparatory Steps

Automotive With CM871

- Remove the turbocharger and inspect for reuse. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Remove the turbocharger actuator. Refer to Procedure 010-134 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-134.html)

Note : This repair requires the use of different turbocharger housing kits, depending on the engine CPL, and turbocharger serial number (found on the turbocharger dataplate). Reference the appropriate parts release information to determine which turbocharger housing kit to use.

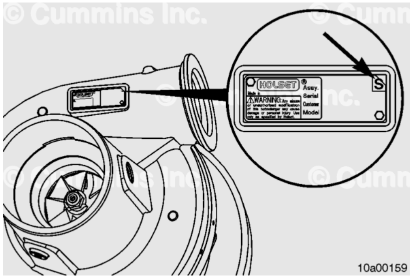
Disassemble

Automotive with CM870

Check for an "S" mark on the top right corner of the turbocharger data tag. This mark indicates that this repair procedure has already been completed on this turbocharger.

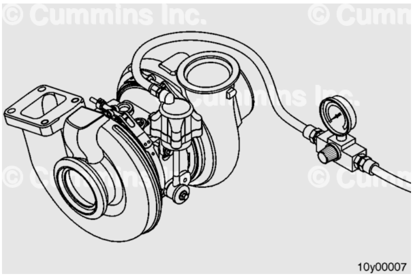
If there is an "S" mark present, do **not** proceed with this repair procedure; proceed with turbocharger replacement.

Cover the oil supply and drain ports in the bearing housing. Use clean care parts, Part Number 4919216 and 4919221.



⚠ WARNING ⚠

Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.

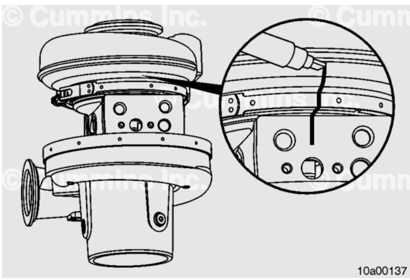


With the actuator in place, use coupling, Part Number 3824843, to connect a regulated air supply to the actuator air inlet. Check the actuator travel. Remove air to allow the actuator to retract. Mark or scribe the VG actuator rod at the base of the actuator. Measure the rod travel by measuring the distance from the base of the turbocharger actuator to the scribe mark. Use a straight edge steel ruler to measure.

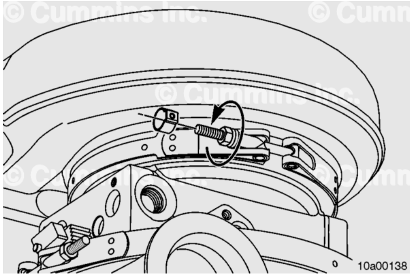
The turbocharger actuator rod should extend more than 10 mm [0.394 in]. If it does **not**, verify the turbocharger actuator is functioning correctly. Refer to Procedure 010-113 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-113-tr.html)If functioning correctly, install to the turbocharger and continue through the procedure.

Create an alignment mark on the turbocharger turbine housing, bearing housing, and V-band clamp in line with the oil drain on the bearing housing.

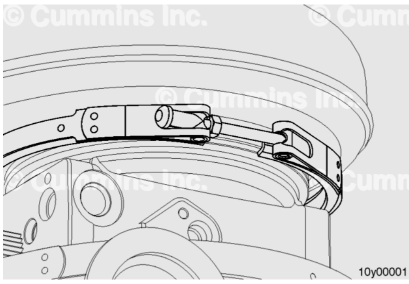
This mark will enable the components to be oriented correctly during assembly.



Loosen the turbocharger turbine housing V-band clamp. The turbine housing V-band clamp locknut cover **must** be removed (use an 11 mm socket) before the locknut can be loosened.

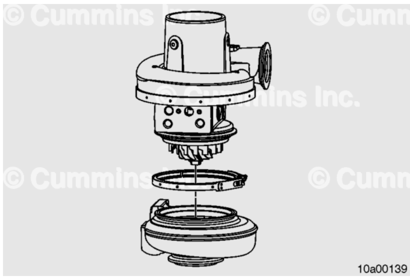


After the clamp locknut is loosened, the V-band clamp can be spread and locked open by using the threaded end of the clamp, pressing it against the opposite end. This can aid in the removal of the turbine housing.



⚠ CAUTION ⚠

The turbine blades or the nozzle ring can be easily damaged. Care is required for the turbine housing removal process.

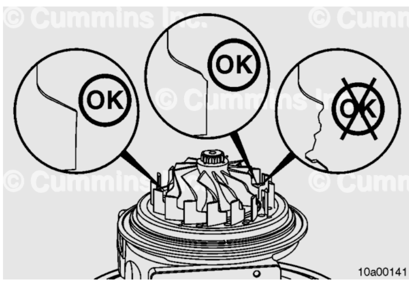


Slightly lift or angle the turbocharger while using a lead hammer to tap the turbine housing down against the bench surface.

Tap so that the turbine housing can be removed squarely. This will reduce the possibility of damage to the nozzle ring assembly. It can take a considerable amount of force to remove the turbine housing.

As the bearing housing and compressor housing assembly loosen, gently lift the assembly from the turbine housing and carefully place the compressor housing side down.

Once the turbocharger turbine housing is removed, inspect the turbine wheel blades for damage.

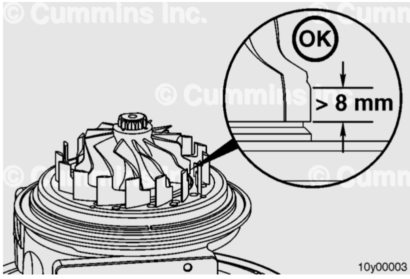


If the blades are damaged, the turbocharger **must** be replaced.

A small amount of wear on the tips is acceptable and will **not** affect the operation of the turbocharger.

Use a straight-edge gauge to measure the length from the bottom-most blade on the turbine wheel to the point where wear begins on the wheel (if any).

This length should be greater than 8 mm [0.315 in]. If it is **not**, the turbocharger **must** be replaced.



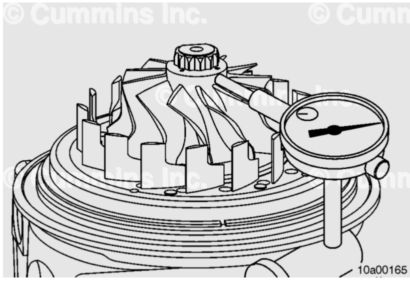
Use a dial indicator to check radial movement at the turbine wheel nose.

Push the turbine wheel away from the gauge and set the indicator to zero.

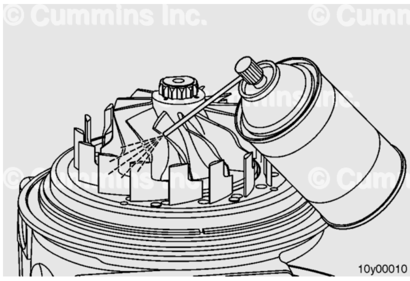
Push the turbine toward the gauge and record the reading.

Turbocharger Radial Bearing Clearance		
mm		**in**
0.590	MAX	0.023

If the turbocharger radial clearance is **not** within specifications, the turbocharger **must** be replaced.

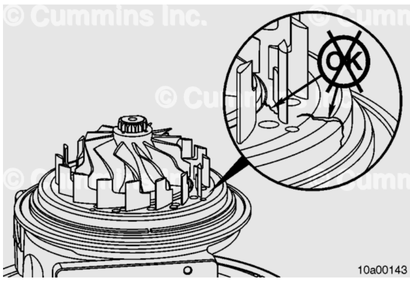


Use contact cleaner, or equivalent, to carefully clean the nozzle ring and vanes.



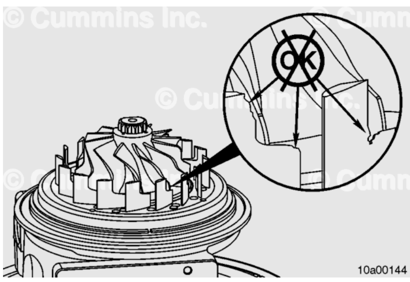
Inspect the variable geometry turbocharger nozzle ring for cracking.

If an axial crack on the nozzle ring is observed, the turbocharger **must** be replaced.



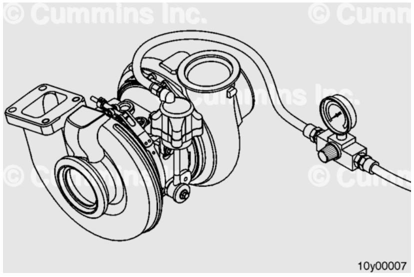
Inspect the variable geometry turbocharger nozzle ring for vane damage.

If there are dents, dings, or pieces broken out of **any** of the nozzle vanes, the turbocharger **must** be replaced.



⚠ WARNING ⚠

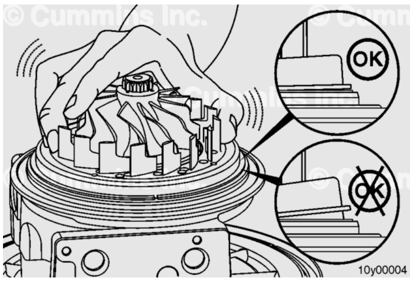
Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.



Use regulated air with the VGT actuator to extend the nozzle ring.

To reduce the spring load at the linkage, use coupling, Part Number 3824843, to apply 414 kPa [60 psi] of air pressure to the actuator air inlet.

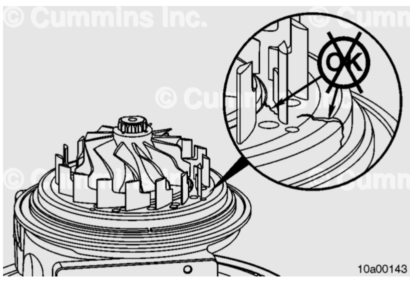
Tilt the nozzle ring back and forth by hand to check for excessive play.



The nozzle should rock back and forth gently, but if one side of the nozzle can drop significantly more than the other, the play is excessive.

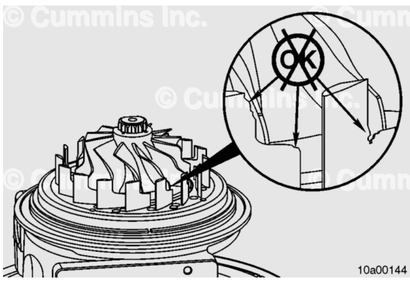
If there is excessive play in the nozzle ring, the turbocharger **must** be replaced.

Inspect the variable geometry turbocharger nozzle ring for cracking.



If an axial crack on the nozzle ring is observed, the turbocharger **must** be replaced.

Inspect the variable geometry turbocharger nozzle ring for vane damage.



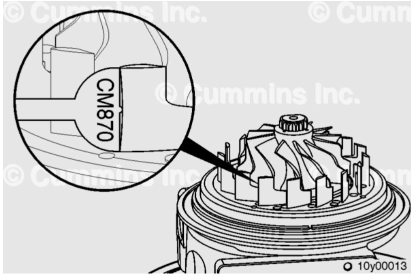
If there are dents, dings, or pieces broken out of **any** of the nozzle vanes, the turbocharger **must** be replaced.

Check for excessive wear on the leading edge of the nozzle vanes.

Use the nozzle vane wear gauge, Part Number 2892218, to check for excessive wear. Use the CM870 or CM871 end of the wear gauge, depending on the turbocharger. Place the nozzle vane wear gauge against each vane and down against the nozzle ring. It is important to place the gauge down completely against the nozzle ring face.

If the gauge can rock back and forth, the wear is **not** excessive. If the gauge can **not** rock back and forth, the nozzle vane has excessive wear.

If excessive wear is found and a vane does **not** pass the wear gauge check, the turbocharger **must** be replaced.



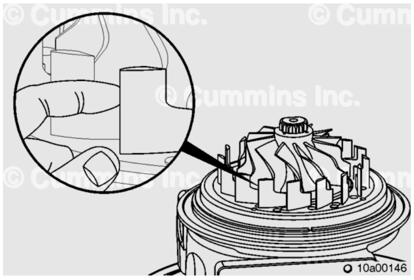
Check for burrs on the leading edge of the nozzle vanes.

The surface of the vane face should be smooth. A burr can be felt with the fingertip.

These burrs are typical and can be removed.

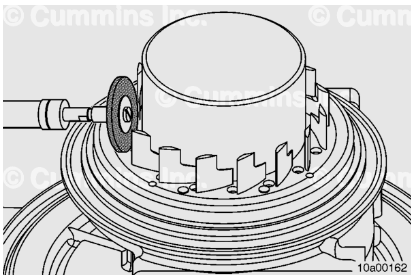
If burrs are present, place the turbine wheel protective cap, Part Number 2892213, over the turbine wheel.

Use an abrasive disc and air tool or drill to remove the burrs from the edge of each vane.



Make sure the burrs are removed completely from each vane by feeling with the fingertip.

Carefully clean off any debris around the bearing housing surface and seal groove.

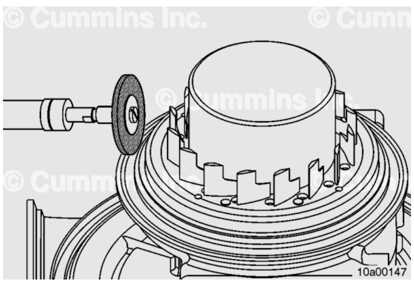


Use a wire brush and air tool or drill to clean the turbine housing mating surface on the bearing housing.

Clean both the horizontal and vertical mating surfaces.

Make sure the groove in which the seal ring sits is completely clean.

After completed, remove the turbine wheel protective cap.

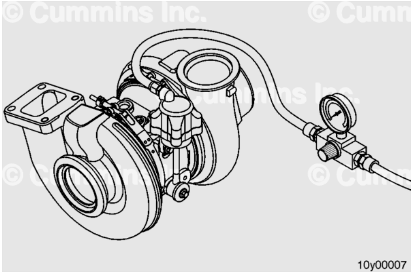


⚠ WARNING ⚠

Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.

Use regulated air with the VGT actuator to extend the nozzle ring.

To reduce the spring load at the linkage, use coupling, Part Number 3824843, to apply 414 kPa [60 psi] of air pressure to the actuator air inlet.

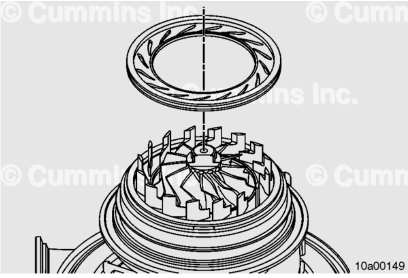


Use the shroud gauge tool, Part Number 2892179, to check the nozzle vanes for bending. It is important to use the correct shroud gauge tool. ISX CM870 and ISX CM871 turbochargers use different shroud gauge tools. Check the part number to make sure the correct one is used.

Carefully align and rotate the shroud gauge tool until all nozzle vanes are engaged. Keep the shroud gauge tool parallel to the table. Slowly move it along the nozzle vanes until the shroud gauge tool is completely against the nozzle ring face.

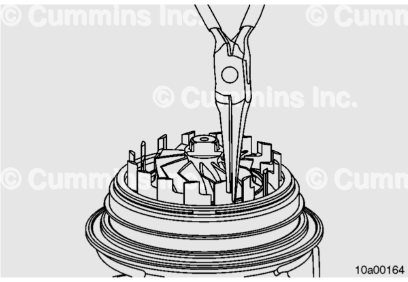
The shroud gauge tool should be able to drop completely to the bottom of the nozzle ring assembly. Slowly pull the shroud plate tool back up and out of the nozzle vanes.

If the shroud gauge tool does **not** move freely, inspect for burrs that were missed on the nozzle vanes and remove, if possible.



If the shroud gauge tool continues to **not** fully engage, or if it requires excessive force to move up and down, bending of the nozzle vanes can possibly be required.

If it does **not** move freely after several attempts of bending a nozzle vane and checking with the shroud gauge tool, the turbocharger **must** be replaced.



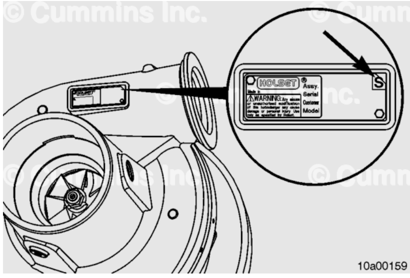
Disassemble

Automotive With CM871

Cover the oil supply and drain ports in the bearing housing.
Use clean care parts, Part Number 4919216 and 4919221.

Check for an "S" mark on the top right corner of the turbocharger data tag. This mark indicates that this repair procedure has already been completed on this turbocharger.

If there is an "S" mark present, do **not** proceed with this repair procedure; proceed with turbocharger replacement.



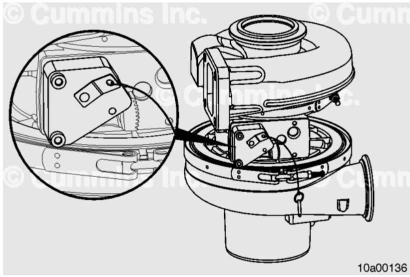
Place the turbocharger on a flat sturdy table with the compressor housing down.

Lock the nozzle ring by using the sector gear positioning tool, Part Number 2892215.

Place the sector gear positioning tool over the sector gear and tighten the capscrews in the bottom left and top right VGT actuator mounting capscrew holes finger tight.

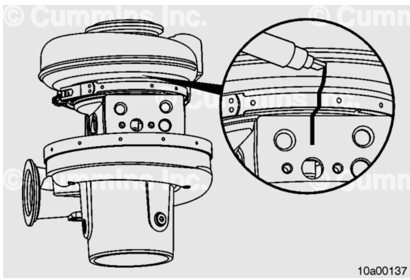
It is **not** necessary to place the locking pin in the sector gear positioning tool at this time.

It is important to have the sector gear positioning tool in place prior to removing the turbine housing in order to reduce the possibility of over-stroking the nozzle ring.

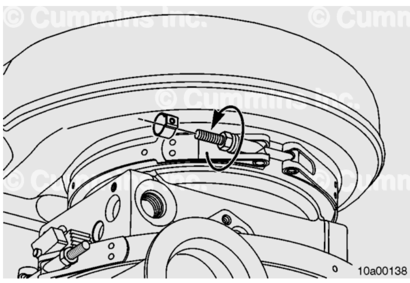


Create an alignment mark on the turbocharger turbine housing, bearing housing, and V-band clamp in line with the oil drain on the bearing housing.

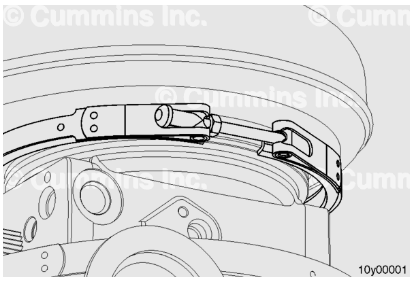
This mark will enable the components to be oriented correctly during assembly.



Loosen the turbocharger turbine housing V-band clamp. The turbine housing V-band clamp locknut cover **must** be removed (use an 11 mm socket) before the locknut can be loosened.

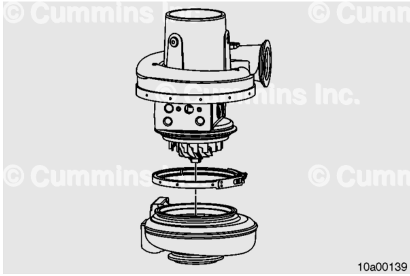


After the clamp locknut is loosened, the V-band clamp can be spread and locked open by using the threaded end of the clamp, pressing it against the opposite end. This can aid in the removal of the turbine housing.



⚠ CAUTION ⚠

The turbine blades or the nozzle ring can be easily damaged. Care is required for the turbine housing removal process.



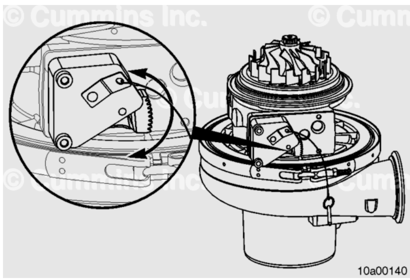
Slightly lift or angle the turbocharger while using a lead hammer to tap the turbine housing down against the bench surface.

Tap so that the turbine housing can be removed squarely. This will reduce the possibility of damage to the nozzle ring assembly. It can take a considerable amount of force to remove the turbine housing.

As the bearing housing and compressor housing assembly loosen, gently lift the assembly from the turbine housing and carefully place the compressor housing side down.

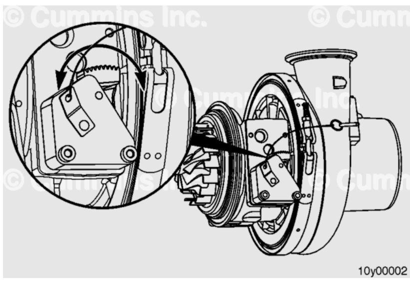
With the turbine housing removed and the sector gear positioning tool, Part Number 2892215, still installed, move the sector gear back and forth by hand to check for smooth movement.

If the sector gear movement is **not** smooth, the turbocharger **must** be replaced.



Carefully rotate the turbocharger on its side and move the sector gear back and forth by hand again to check for smooth movement.

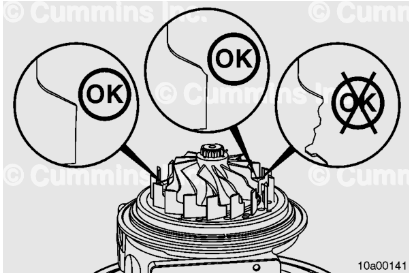
If the sector gear movement is **not** smooth, the turbocharger **must** be replaced.



Once the turbocharger turbine housing is removed, inspect the turbine wheel blades for damage.

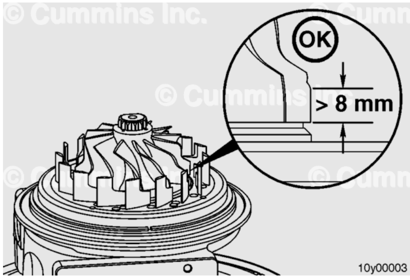
If the blades are damaged, the turbocharger **must** be replaced.

A small amount of wear on the tips is acceptable and will **not** affect the operation of the turbocharger.



Use a straightedge gauge to measure the length from the bottom-most blade on the turbine wheel to the point where wear begins on the wheel (if any).

This length should be greater than 8 mm [0.315 in]. If it is **not**, the turbocharger **must** be replaced.



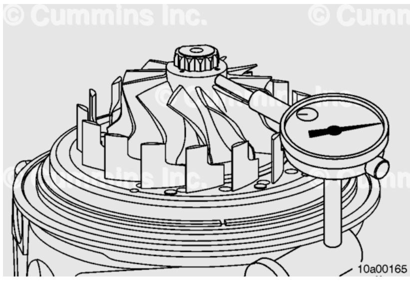
Use a dial indicator to check radial movement at the turbine wheel nose.

Push the turbine wheel away from the gauge and set the indicator to zero.

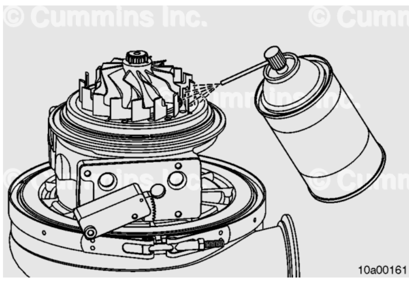
Push the turbine toward the gauge and record the reading.

Turbocharger Radial Bearing Clearance		
mm		**in**
0.590	MAX	0.023

If the turbocharger radial clearance is **not** within specifications, the turbocharger **must** be replaced.

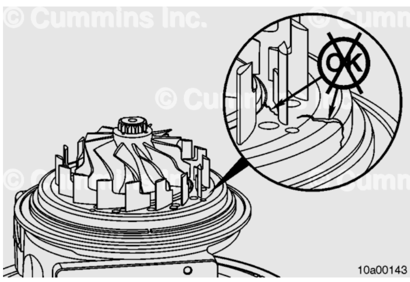


Use contact cleaner, or equivalent, to carefully clean the nozzle ring and vanes.



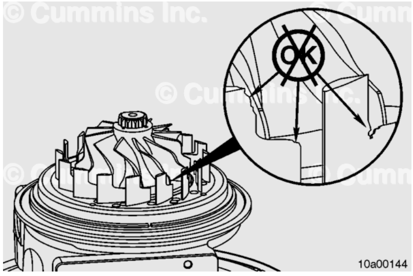
Inspect the variable geometry turbocharger nozzle ring for cracking.

If an axial crack on the nozzle ring is observed, the turbocharger **must** be replaced.



Inspect the variable geometry turbocharger nozzle ring for vane damage.

If there are dents, dings, or pieces broken out of **any** of the nozzle vanes, the turbocharger **must** be replaced.



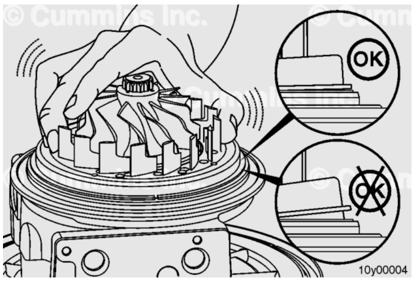
Extend the nozzle ring out and lock it in place by placing the locking pin through the sector gear positioning tool and into the sector gear hole.

Tilt the nozzle ring back and forth by hand to check for excessive play.

The nozzle should rock bach and forth slightly. If one side of the nozzle can drop significantly more than the other, the play is excessive.

If there is excessive play in the nozzle ring, the turbocharger **must** be replaced.

After completed, the pin going through the sector gear positioning tool and sector gear hole can be removed.

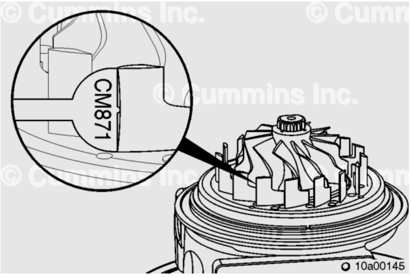


Check for excessive wear on the leading edge of the nozzle vanes.

Use the nozzle vane wear gauge, Part Number 2892212, to check for excessive wear. Use the CM870 or CM871 end of the wear gauge, depending on the turbocharger. Place the nozzle vane wear gauge against each vane and down against the nozzle ring. It is important to place the gauge down completely against the nozzle ring face.

If the gauge can rock back and forth, the wear is **not** excessive. If the gauge can **not** rock back and forth, the nozzle vane has excessive wear.

If excessive wear is found and a vane does **not** pass the wear gauge check, the turbocharger **must** be replaced.

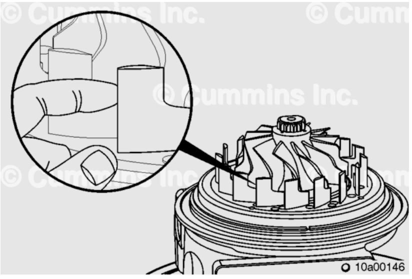


Check for burrs on the leading edge of the nozzle vanes.

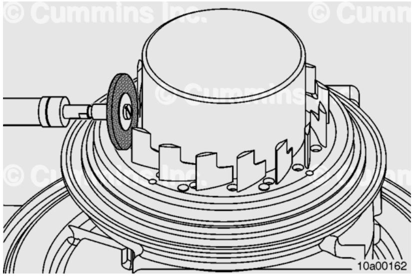
The surface of the vane face should be smooth. A burr can be felt with the fingertip.

These burrs are typical and can be removed.

If burrs are present, place the turbine wheel protective cap, Part Number 2892213, over the turbine wheel.



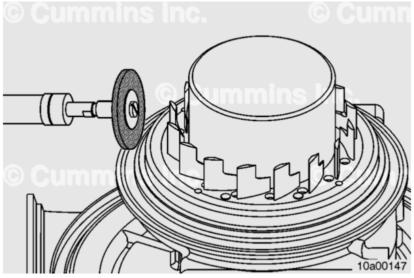
Use an abrasive disc, Part Number 3824541, and air tool or drill to clean up the burrs on the edge of each vane. Do **not** use excessive force or time to clean the vanes; burrs are fairly easy to remove.



Make sure the burrs are removed completely from each vane by feeling with the fingertip.

Carefully clean off any debris around the bearing housing surface and seal groove.

Use a wire brush and air tool or drill to clean the turbine housing mating surface on the bearing housing.

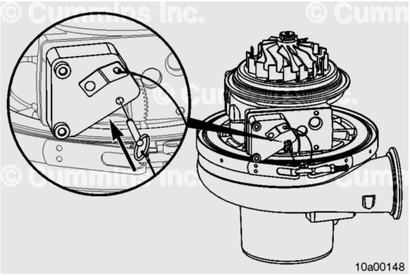


Clean both the horizontal and vertical mating surfaces.

Make sure the groove in which the seal ring sits is completely clean.

After completed, remove the turbine wheel protective cap.

Use the sector gear positioning tool, Part Number 2892215, to lock the nozzle ring in the fully open position.



Place the sector gear positioning tool over the sector gear and tighten the capscrews in the bottom left and top right VGT actuator mounting capscrew holes hand-tight.

Insert the locking pin through the sector gear positioning tool and into the sector gear hole.

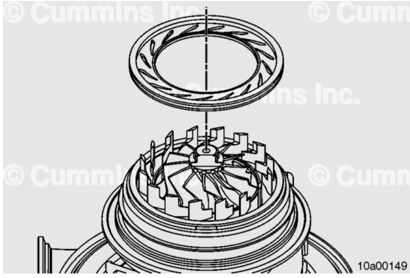
At this point, the nozzle ring should be at the fully extended position.

Use the shroud gauge tool, Part Number 2892180, to check the nozzle vanes for bending. It is important to use the correct shroud gauge tool. ISX CM870 and ISX CM871 turbochargers use different shroud gauge tools. Check the part number to make sure the correct one is used.

Carefully align and rotate the shroud gauge tool until all nozzle vanes are engaged. Keep the shroud gauge tool parallel to the table. Slowly move it along the nozzle vanes until the shroud gauge tool is completely against the nozzle ring face.

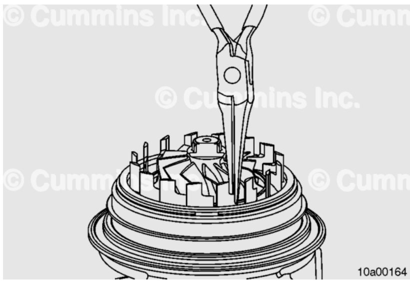
The shroud gauge tool should be able to drop completely to the bottom of the nozzle ring assembly. Slowly pull the shroud plate tool back up and out of the nozzle vanes.

If the shroud gauge tool does **not** move freely, inspect for burrs that were missed on the nozzle vanes and remove, if possible.



If the shroud gauge tool continues to **not** fully engage, or if it requires excessive force to move up and down, bending of the nozzle vanes can possibly be required.

If it does **not** move freely after several attempts of bending a nozzle vane and checking with the shroud gauge tool, the turbocharger **must** be replaced.

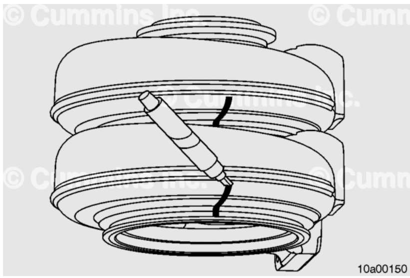


Assemble

Automotive with CM870

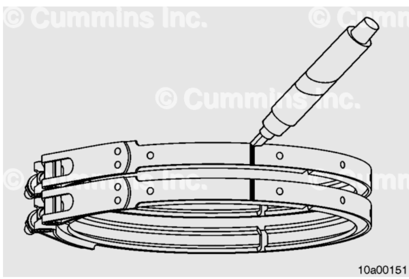
Place the new turbine housing assembly directly on top of the old turbine housing. Align the mounting flanges so that they are parallel.

Use the alignment marks that were made on the old turbine housing to the new turbine housing so that it can be aligned properly to the turbocharger.



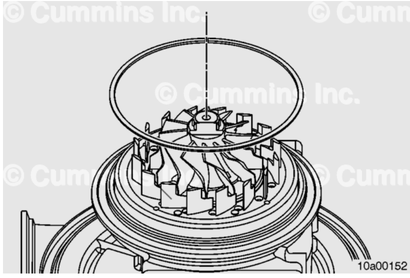
Place the new V-band clamp directly on top of the old V-band clamp. Align them exactly.

Use the alignment marks that were made on the old V-band clamp to the new V-band clamp so that it can be properly aligned to the turbocharger.

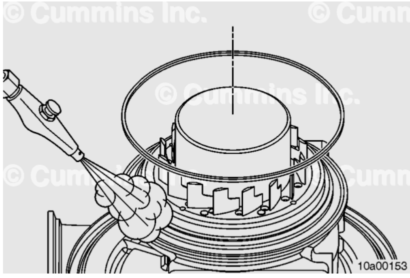


Remove and replace the seal ring that goes around the nozzle ring.

The gap for the seal ring can be placed in any position.

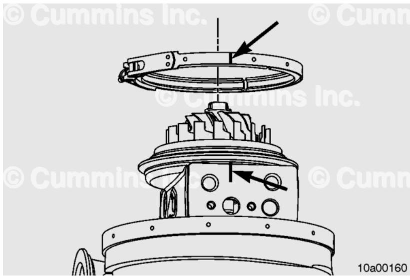


- With the turbine wheel protective cap, Part Number 2892213, on the turbine wheel, carefully blow off any debris in the groove where the C-seal sits.
- Place the new C-seal (bearing housing to turbine housing seal) onto the sealing surface of the turbocharger bearing housing.
- Anti-seize compound can be placed in a few places around the circumference of the C-seal to better hold it in place during the assembly process.

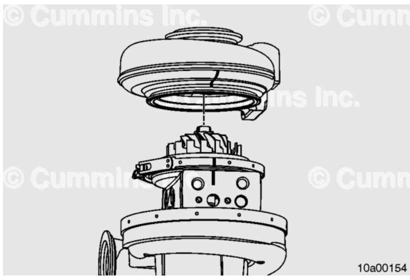


- Replace the turbine housing V-band clamp.
- To install the turbine housing, position the V-band clamp over the bearing housing and align the ink marks applied during the disassembly process.

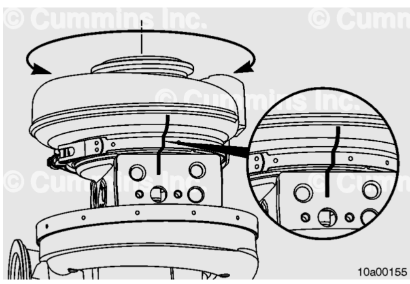
The V-band clamp can be spread and locked open by pressing the threaded end of the clamp against the opposite end. This will aid in allowing the turbine housing to drop into place.



- With the nozzle ring still at an extended position, carefully lower the turbine housing into the bearing and compressor housing assembly until the nozzle vanes contact the shroud plate.
- Carefully rotate the turbine housing, as needed, in order to engage all of the nozzle vanes.
- Remain as parallel to the table as possible while lowering the turbine housing, until the nozzle vanes are fully engaged.

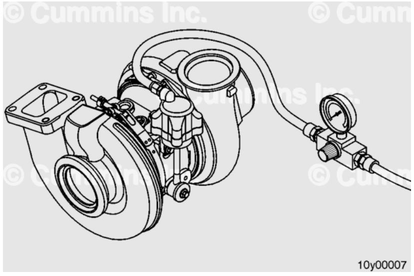


- Carefully rotate the turbine housing until the marks on the bearing housing and turbine housing are directly aligned.



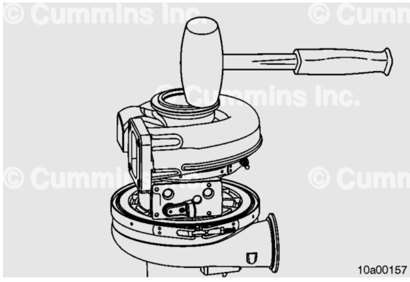
⚠ WARNING ⚠

Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.



Release the actuator by disconnecting the coupling and regulated air supply.

Carefully tap on the turbine housing with a lead hammer until the turbine housing is fully seated on the bearing housing surface.



Tighten the locknut on the V-band clamp.

Place the new V-band clamp in the correct orientation and tighten the locknut.

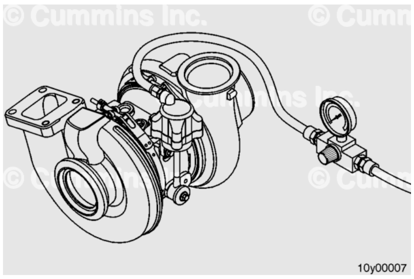
Torque Value: 11.3 n•m [100 in-lb]

Loosen the locknut 180 degrees and tighten the locknut.

Torque Value: 11.3 n•m [100 in-lb]

⚠ WARNING ⚠

Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.



Use coupling, Part Number 3824843, to connect a regulated air supply to the actuator air inlet. Check the actuator travel. Remove air to allow the actuator to retract. Use a straight edge steel ruler to measure.

The turbocharger actuator rod should extend more than 10 mm [0.394 in].

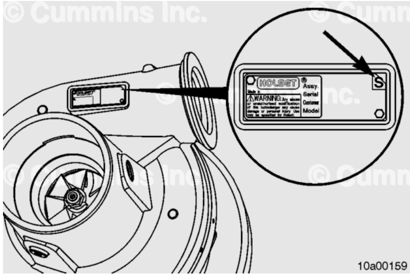
If movement is **not** smooth, or if there is **not** full travel, rotate the turbine housing **clockwise**. Rotate **counterclockwise** back to the alignment marks.

Check for full travel of the actuator rod.

Several rotational movements can need to be made in order to correctly align the nozzle ring and the sector gear.

Use a metal stamping tool to mark an “S” on the top right corner of the turbocharger data tag.

It is important that it be marked in the corner, as the middle of the data tag may **not** have a surface behind it.

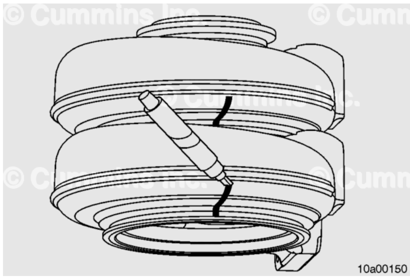


Assemble

Automotive With CM871

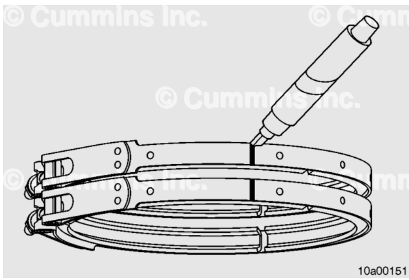
Place the new turbine housing assembly directly on top of the old turbine housing. Align the mounting flanges so that they are parallel.

Use the alignment marks that were made on the old turbine housing to the new turbine housing so that it can be aligned properly to the turbocharger.



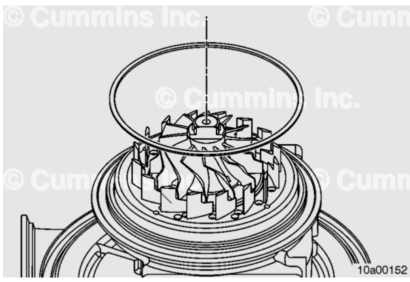
Place the new V-band clamp directly on top of the old V-band clamp. Align them exactly.

Use the alignment marks that were made on the old V-band clamp to the new V-band clamp so that it can be properly aligned to the turbocharger.



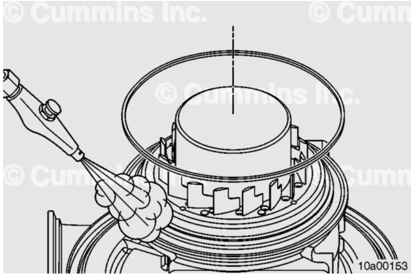
Remove and replace the seal ring that goes around the nozzle ring.

The gap for the seal ring can be placed in any position.



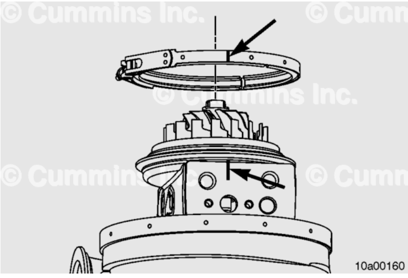
- With the turbine wheel protective cap, Part Number 2892213, on the turbine wheel, carefully blow off any debris in the groove where the C-seal sits.
- Place the new C-seal (bearing housing to turbine housing seal) onto the sealing surface of the turbocharger bearing housing.

- Anti-seize compound can be placed in a few places around the circumference of the C-seal to better hold it in place during the assembly process.

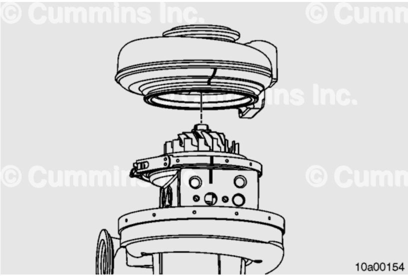


- Replace the turbine housing V-band clamp.
- To install the turbine housing, position the V-band clamp over the bearing housing and align the ink marks applied during the disassembly process.

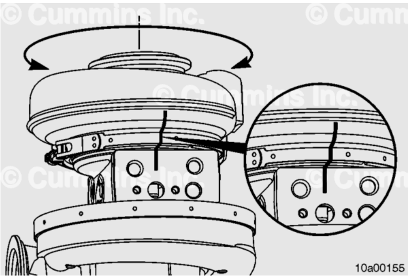
The V-band clamp can be spread and locked open by pressing the threaded end of the clamp against the opposite end. This will aid in allowing the turbine housing to drop into place.



- Make sure the sector gear positioning tool is still installed and the nozzle ring is at the fully extended position (the locking pin should be installed).
- Carefully lower the turbine housing into the bearing and compressor housing assembly until the nozzle vanes contact the shroud plate.
- Carefully rotate the turbine housing, as needed, in order to engage all of the nozzle vanes.
- Remain as parallel to the table as possible while lowering the turbine housing, until the nozzle vanes are fully engaged.

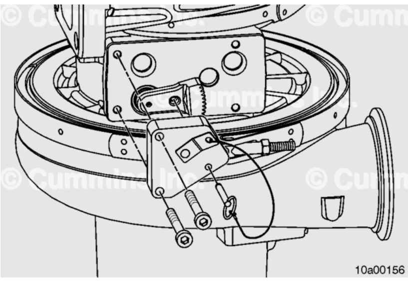


- Carefully rotate the turbine housing until the marks on the bearing housing and turbine housing are directly aligned.

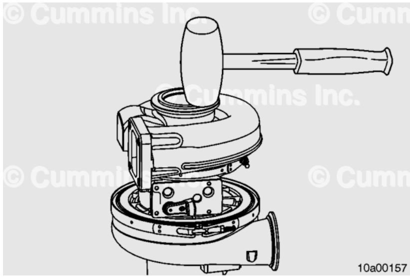


Carefully remove the sector gear positioning tool, Part Number 2892215.

Note : If the turbine housing has to be removed at any point, the sector gear positioning tool must be installed first.



Carefully tap on the turbine housing with a lead hammer until the turbine housing is fully seated on the bearing housing surface.



Tighten the locknut on the V-band clamp.

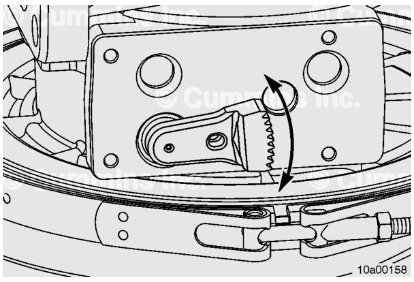
Place the new V-band clamp in the correct orientation and tighten the locknut.

Torque Value: 11.3 n•m [100 in-lb]

Loosen the locknut 180 degrees and tighten the locknut.

Torque Value: 11.3 n•m [100 in-lb]

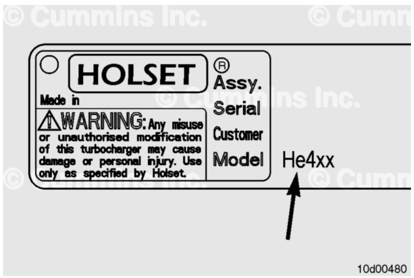
Move the sector gear back and forth by hand to check for smooth movement.



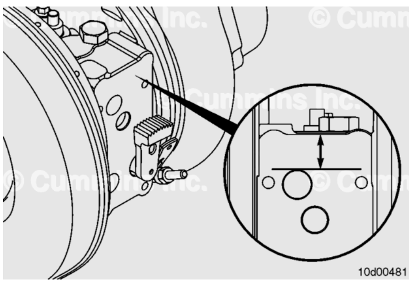
If movement is **not** smooth, rotate the turbine housing **clockwise**. Then, rotate **counterclockwise** back to the alignment marks.

Check for smooth movement of the sector gear again. Several rotational movements can need to be made in order to correctly align the nozzle ring and the sector gear.

The sector gear travel gauges are designed for specific turbocharger models. To determine the correct gauge, verify the turbocharger model number from the turbocharger dataplate. Select the gauge that matches the first three letters of the turbocharger model number.



If the turbocharger data tag is missing or covered with paint, an alternate method of determining the turbocharger model number is to measure the distance between the top of the cup plug hole to the bearing housing edge.



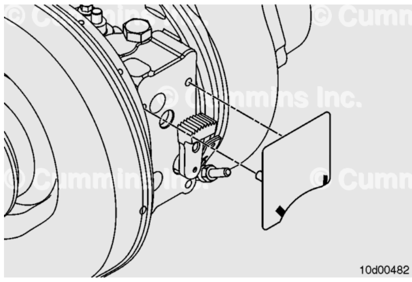
If the distance is greater than 19 mm [0.748 in], the turbocharger model is HE4xx.

If the distance is less than 19 mm [0.748 in], the model is HE5xx.

Carefully bend the gauge and slide the thin section under the sector gear to install the sector gear travel gauge.

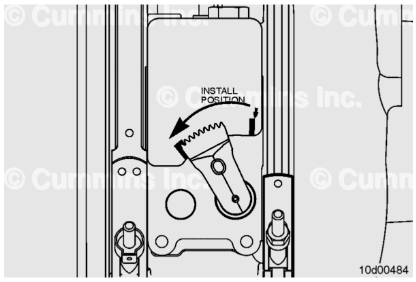
If necessary, pull the sector gear out by hand to allow more clearance for the gauge.

Verify the three alignment bosses are fully engaged in the bearing housing.



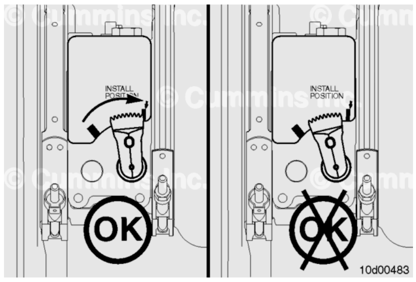
Rotate the sector gear **counterclockwise**, toward the turbocharger turbine housing, until it stops.

The edge of the sector gear **must** be in the green acceptance zone of the gauge.



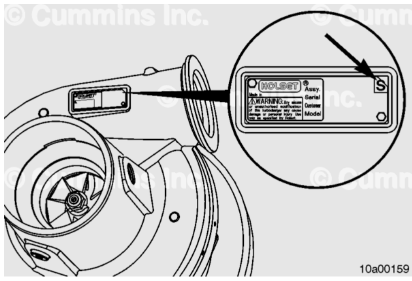
Rotate the sector gear **clockwise**, toward the turbocharger compressor housing, until it stops. The edge of the sector gear **must** be in the green acceptance zone of the gauge.

If the sector gear does **not** go through the full range of motion, or if the sector gear requires excessive force to move it by hand, rotate the turbine housing back and forth until a smooth and full motion of the sector gear is achieved.



Use a metal stamping tool to mark an “S” on the top right corner of the turbocharger data tag.

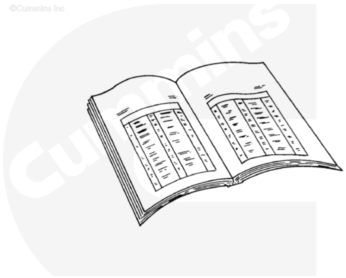
It is important that it be marked in the corner, as the middle of the data tag may **not** have a surface behind it.



Finishing Steps

Automotive with CM870

- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Operate the engine and check for leaks.



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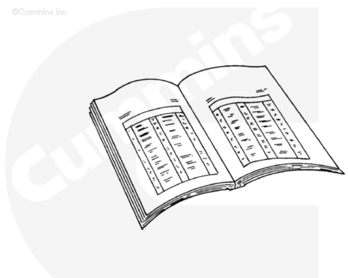
Finishing Steps

Automotive With CM871

- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Install the turbocharger actuator. Refer to Procedure 010-134 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-134.html)
- Operate the engine and check for leaks.

If a malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013.html)



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